

Question number	Answer	Additional guidance	Mark
6(a)	An explanation that includes three of the following points: • (less cholesterol) so reduces {build up of cholesterol / formation of atheroma / formation of a plaque} (1) • in the (walls of the) coronary artery (1) • so flow of blood to heart {muscle / cells / tissue} would not be prevented (1) • so oxygen available for respiration for heart {muscle / cells / tissue / contraction} (1)	NB Award converses but must be in context of less cholesterol for full marks to be awarded	(3)

Question number	Answer	Additional guidance	Mark
6(b)	Any two of the following:		

	• (total) number of amino acids (in a peptide) / length of {(poly)peptide / chain} • {sequence / order / position} of amino acids • {proportion / number} of each type of amino acids (1)	ACCEPT {sequence / order / position} R groups ACCEPT (types of) amino acids present R groups for amino acids	(1)
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6(c)(i)	An explanation that includes two of the following points: • because this (digestion) happens when a person eats soybeans / scientists wanted to mimic what was happening in the body (1) • to {extract / separate / release} the {proteins / glycinin and β-conglycinin / active ingredients} (1) • as enzymes needed to break (peptide) bonds (to produce the peptides) (1)	ACCEPT copies digestion ACCEPT so that the proteins are soluble	(2)

Question number	Answer	Additional guidance	Mark
6(c)(ii)	• because they (different enzymes) are {found in different parts of the digestive system / needed to break down different proteins} (1)	ACCEPT more than one type of enzyme in the body two or more named parts of digestive system different substrates	(1)

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Question number	Answer	Additional guidance	Mark
6(c)(iii)	An explanation that includes the following points: • 37°C because this is the {optimum temperature (for human enzymes) / (human) body temperature} (1) • 2 hours because {this is enough time for complete digestion to occur / this is how long the digesting food stays in each part of the digestive system / this allows all the substrate to be broken down} (1)	ACCEPT time spent in stomach / (small) intestine sufficient time for enough peptides to be produced for the investigation	(2)

Question number	Answer	Additional guidance	Mark
6(d)	An explanation that includes three of the following points:		(3)

	• as the concentration of soybean peptides increase, the inhibition of HMGCR increases and the synthesis of cholesterol decreases (1) • because soybean peptides inhibit HMGCR • because the peptides {bind to the HMGCR / compete with the substrate} (1) • as the cholesterol precursor cannot be converted into cholesterol (by HMGCR) (1)	NB Piece together ACCEPT bind to the active site ACCEPT cholesterol precursor cannot bind to HMGCR	
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